

## Lab No. 2

2024-03-26

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(ggplot2)
library(critstats)
```

### 1.1

I am continuing with using the Public School Characteristics 2022-23 data set conducted by the National Center for Education Statistics (NCES). The NCES gathers demographic and geographic data about U.S. public schools and factors such as enrollment, percentage of students with free or reduced lunch eligibility, and location.

### 1.2

This lab is a multivariable analysis focusing on free/reduced lunch rates, student-to-teacher ratio, locale of the school, and race of each student.

### 1.3

What is the relationship between the percentage of students eligible for free/reduced-price lunch and class sizes in traditional U.S. public schools, and how do these relationships vary by the racial composition of the student body and school geography?

## 1.4

The first step is to clean the data, using Lab #1 as a starting off point of which columns to drop and renaming columns to increase readability. However, for this lab, I will include columns ‘School Type’ and ‘Status’ to limit the schools that will be analyzed to regular operational schools for my main dataframe. An additional dataframe will be created that will include all operational schools, which includes alternative, career & technical, and special education schools. To do this, I will first find the unique values for the school type, status, and virtual schools columns, so that I can create the two dataframes. Additional steps will be taken to create another column in each dataframe which contains values of the exact percentage of students receiving free/reduced lunch at each school. Lastly, I will take care of any missing or incorrect values by removing them from the dataframe. The next step will involve exploring the data through descriptive statistics, chi-square tests, and visualizations. For visualizations, I plan to use faceted scatter plots to show how the correlation between lunch rates and student-to-teacher ratios varies between schools depending on the predominant race of students attending each school, and depending on school locale.

## 1.5

There are several potential limitations for this analysis. The first is that the student-to-teacher ratio might be extreme when looking at non-traditional schools, which could skew the results. The solution to this is by analyzing two different dataframes – one that includes regular schools, and one that includes all schools. Additionally, for this analysis, school funding and resource allocation are represented by school lunch rates and student-to-teacher ratios respectively. However, there could be factors that affect these variables that are not seen in this analysis.

## 1.6

As indicated by the authors of the dataset, missing values are represented by a number of indicators, specifically for columns regarding free/reduced lunch: -1 indicates that data is missing, -2 or N indicates that data is not applicable, and -9 indicates that data did not meet NCES data quality standards. These values will be dealt with by removing the schools where these values are present.

## 1.7

### Cleaning the Data

#### Uploading Dataset

```
PSC <- read_csv("/Users/camrynadavis/Desktop/datasets/Public_School_Characteristics_2022-23.csv")
```

```
## Rows: 101390 Columns: 77
## -- Column specification -----
## Delimiter: ","
## chr (23): NCESSCH, SURVYEAR, STABR, LEAID, ST_LEAID, LEA_NAME, SCH_NAME, LST...
## dbl (54): X, Y, OBJECTID, STATUS, TOTFRL, FRELCH, REDLCH, DIRECTCERT, PK, KG...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

## PSC

```
## # A tibble: 101,390 x 77
##       X      Y OBJECTID NCESSCH SURVYEAR STABR LEAID ST_LEAID LEA_NAME SCH_NAME
##   <dbl> <dbl>   <dbl> <chr>   <chr>   <chr> <chr> <chr>   <chr>   <chr>
## 1 -86.2  34.3     1 0100005~ 2022-20~ AL    0100~ AL-101  Albertv~ Albertv~
## 2 -86.2  34.3     2 0100005~ 2022-20~ AL    0100~ AL-101  Albertv~ Albertv~
## 3 -86.2  34.3     3 0100005~ 2022-20~ AL    0100~ AL-101  Albertv~ Albertv~
## 4 -86.2  34.3     4 0100005~ 2022-20~ AL    0100~ AL-101  Albertv~ Albertv~
## 5 -86.2  34.3     5 0100005~ 2022-20~ AL    0100~ AL-101  Albertv~ Albertv~
## 6 -86.2  34.3     6 0100005~ 2022-20~ AL    0100~ AL-101  Albertv~ Albertv~
## 7 -86.3  34.5     7 0100006~ 2022-20~ AL    0100~ AL-048  Marshal~ Kate Du~
## 8 -86.1  34.4     8 0100006~ 2022-20~ AL    0100~ AL-048  Marshal~ Asbury ~
## 9 -86.3  34.4     9 0100006~ 2022-20~ AL    0100~ AL-048  Marshal~ Claysvi~
## 10 -86.3 34.2    10 0100006~ 2022-20~ AL    0100~ AL-048  Marshal~ Douglas~
## # i 101,380 more rows
## # i 67 more variables: LSTREET1 <chr>, LSTREET2 <chr>, LCITY <chr>,
## #   LSTATE <chr>, LZIP <chr>, LZIP4 <chr>, PHONE <chr>, CHARTER_TEXT <chr>,
## #   VIRTUAL <chr>, GSLO <chr>, GSHI <chr>, SCHOOL_LEVEL <chr>, STATUS <dbl>,
## #   SCHOOL_TYPE_TEXT <chr>, SY_STATUS_TEXT <chr>, ULOCALE <chr>, NMCNTY <chr>,
## #   TOTFRL <dbl>, FRELCH <dbl>, REDLCH <dbl>, DIRECTCERT <dbl>, PK <dbl>,
## #   KG <dbl>, G01 <dbl>, G02 <dbl>, G03 <dbl>, G04 <dbl>, G05 <dbl>, ...
```

## Drop & Rename Columns

```
df0 <- PSC %>%
  select(-OBJECTID, -NCESSCH, -SURVYEAR, -LEA_NAME, -MEMBER, -FTE, -PHONE, -AE, -GSLO, -GSHI, -ST_LEAID,
         rename(StateABR = STABR, SchoolName = SCH_NAME, Street1 = LSTREET1, Street2 = LSTREET2, City = LCITY,
                Zip4 = LZIP4, Charter = CHARTER_TEXT, Virtual = VIRTUAL, SchoolLevel = SCHOOL_LEVEL, Locale = ULOCALE,
                TotalFreeLunch = TOTFRL, FreeLunch = FRELCH, ReducedLunch = REDLCH, MealProgramCertified = DIRLUNCH,
                Grade1 = G01, Grade2 = G02, Grade3 = G03, Grade4 = G04, Grade5 = G05, Grade6 = G06, Grade7 = G07,
                Grade10 = G10, Grade11 = G11, Grade12 = G12, Grade13 = G13, Ungrade = UG, MaleEnrollment = TOTM,
                TotalEnrollment = TOTAL, StudentTeacherRatio = STUTERATIO, AIANMale = AMALM, AIANFemale = AMALF,
                AsianFemale = ASALF, AsianTotal = AS, BlackMale = BLALM, BlackFemale = BLALF, BlackTotal = BL,
                HPITotal = HP, HispanicMale = HIALM, HispanicFemale = HIALF, HispanicTotal = HI, TRMale = TRALM,
                WhiteMale = WHALM, WhiteFemale = WHALF, WhiteTotal = WH, Latitude = LATCOD, Longitude = LONCOD)
df0
```

```
## # A tibble: 101,390 x 62
##   StateABR SchoolName Street1 Street2 City State Zip Zip4 Charter Virtual
##   <chr>      <chr>      <chr>   <chr>   <chr> <chr> <chr> <chr>   <chr>   <chr>
## 1 AL      Albertville~ 600 E ~ <NA>   Albe~ AL   35950 <NA> No      Not Vi~
## 2 AL      Albertville~ 402 E ~ <NA>   Albe~ AL   35950 2322 No      Not Vi~
## 3 AL      Albertville~ 901 W ~ <NA>   Albe~ AL   35950 1300 No      Not Vi~
## 4 AL      Albertville~ 145 We~ <NA>   Albe~ AL   35950 <NA> No      Not Vi~
## 5 AL      Albertville~ 257 Co~ <NA>   Albe~ AL   35951 3927 No      Not Vi~
## 6 AL      Albertville~ 1100 H~ <NA>   Albe~ AL   35950 2532 No      Not Vi~
## 7 AL      Kate Duncan~ 6077 M~ <NA>   Grant AL   35747 <NA> No      Not Vi~
## 8 AL      Asbury High~ 1990 A~ <NA>   Albe~ AL   35951 <NA> No      Not Vi~
## 9 AL      Claysville ~ 140 Cl~ <NA>   Gunt~ AL   35976 <NA> No      Not Vi~
```

```
## 10 AL Douglas Ele~ Highwa~ <NA> Doug~ AL 35964 <NA> No Not Vi~
## # i 101,380 more rows
## # i 52 more variables: SchoolLevel <chr>, SchoolType <chr>, Status <chr>,
## # Locale <chr>, County <chr>, TotalFreeLunch <dbl>, FreeLunch <dbl>,
## # ReducedLunch <dbl>, MealProgramCertified <dbl>, PreK <dbl>,
## # Kindergarten <dbl>, Grade1 <dbl>, Grade2 <dbl>, Grade3 <dbl>, Grade4 <dbl>,
## # Grade5 <dbl>, Grade6 <dbl>, Grade7 <dbl>, Grade8 <dbl>, Grade9 <dbl>,
## # Grade10 <dbl>, Grade11 <dbl>, Grade12 <dbl>, Grade13 <dbl>, ...
```

## Finding Unique Values

```
unique(df0$SchoolType)
```

```
## [1] "Regular School" "Alternative Education School"
## [3] "Career and Technical School" "Special Education School"
```

```
unique(df0$Status)
```

```
## [1] "Currently operational"
## [2] "School to be operational within two years"
## [3] "New school"
## [4] "School temporarily closed"
## [5] "School was operational but not reported last year"
## [6] "School has changed agency"
## [7] "School has reopened"
```

```
unique(df0$Virtual)
```

```
## [1] "Not Virtual" "Not Reported"
## [3] "Full Virtual" "Supplemental Virtual"
## [5] "Virtual with face to face options" "Missing"
```

## Only Include Regular Operational Schools

```
df0 <- df0 %>%
  filter(Virtual == "Not Virtual" & SchoolType == "Regular School"
         & Status != "School to be operational within two years"
         & Status != "School temporarily closed")
```

```
df0
```

```
## # A tibble: 62,155 x 62
## StateABR SchoolName Street1 Street2 City State Zip Zip4 Charter Virtual
## <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 AL Albertville~ 600 E ~ <NA> Albe~ AL 35950 <NA> No Not Vi~
## 2 AL Albertville~ 402 E ~ <NA> Albe~ AL 35950 2322 No Not Vi~
## 3 AL Albertville~ 901 W ~ <NA> Albe~ AL 35950 1300 No Not Vi~
## 4 AL Albertville~ 145 We~ <NA> Albe~ AL 35950 <NA> No Not Vi~
```

```
## 5 AL      Albertville~ 257 Co~ <NA>    Albe~ AL      35951 3927 No      Not Vi~
## 6 AL      Albertville~ 1100 H~ <NA>    Albe~ AL      35950 2532 No      Not Vi~
## 7 AL      Kate Duncan~ 6077 M~ <NA>    Grant AL      35747 <NA> No      Not Vi~
## 8 AL      Asbury High~ 1990 A~ <NA>    Albe~ AL      35951 <NA> No      Not Vi~
## 9 AL      Claysville ~ 140 Cl~ <NA>    Gunt~ AL      35976 <NA> No      Not Vi~
## 10 AL     Douglas Ele~ Highwa~ <NA>    Doug~ AL      35964 <NA> No      Not Vi~
## # i 62,145 more rows
## # i 52 more variables: SchoolLevel <chr>, SchoolType <chr>, Status <chr>,
## #   Locale <chr>, County <chr>, TotalFreeLunch <dbl>, FreeLunch <dbl>,
## #   ReducedLunch <dbl>, MealProgramCertified <dbl>, PreK <dbl>,
## #   Kindergarten <dbl>, Grade1 <dbl>, Grade2 <dbl>, Grade3 <dbl>, Grade4 <dbl>,
## #   Grade5 <dbl>, Grade6 <dbl>, Grade7 <dbl>, Grade8 <dbl>, Grade9 <dbl>,
## #   Grade10 <dbl>, Grade11 <dbl>, Grade12 <dbl>, Grade13 <dbl>, ...
```

## Creating Alt Dataframe (All Operational Schools)

```
df_all <- PSC %>%
  select(-OBJECTID, -NCESSCH, -SURVYEAR, -LEA_NAME, -MEMBER, -FTE, -PHONE, -AE, -GSLO, -GSHI, -ST_LEA
  rename(StateABR = STABR, SchoolName = SCH_NAME, Street1 = LSTREET1, Street2 = LSTREET2, City = LCITY,
  Zip4 = LZIP4, Charter = CHARTER_TEXT, Virtual = VIRTUAL, SchoolLevel = SCHOOL_LEVEL, Locale = U
  TotalFreeLunch = TOTFRL, FreeLunch = FRELCH, ReducedLunch = REDLCH, MealProgramCertified = DIRI
  Grade1 = G01, Grade2 = G02, Grade3 = G03, Grade4 = G04, Grade5 = G05, Grade6 = G06, Grade7 = G
  Grade10 = G10, Grade11 = G11, Grade12 = G12, Grade13 = G13, Ungrade = UG, MaleEnrollment = TOT
  TotalEnrollment = TOTAL, StudentTeacherRatio = STUTERATIO, AIANMale = AMALM, AIANFemale = AMAL
  AsianFemale = ASALF, AsianTotal = AS, BlackMale = BLALM, BlackFemale = BLALF, BlackTotal = BL,
  HPITotal = HP, HispanicMale = HIALM, HispanicFemale = HIALF, HispanicTotal = HI, TRMale = TRAL
  WhiteMale = WHALM, WhiteFemale = WHALF, WhiteTotal = WH, Latitude = LATCOD, Longitude = LONCOD

df_all <- df_all %>%
  filter(Status != "School to be operational within two years" & Status != "School temporarily closed")

df_all
```

```
## # A tibble: 100,571 x 62
##   StateABR SchoolName Street1 Street2 City State Zip Zip4 Charter Virtual
##   <chr>      <chr>      <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 AL      Albertville~ 600 E ~ <NA>    Albe~ AL      35950 <NA> No      Not Vi~
## 2 AL      Albertville~ 402 E ~ <NA>    Albe~ AL      35950 2322 No      Not Vi~
## 3 AL      Albertville~ 901 W ~ <NA>    Albe~ AL      35950 1300 No      Not Vi~
## 4 AL      Albertville~ 145 We~ <NA>    Albe~ AL      35950 <NA> No      Not Vi~
## 5 AL      Albertville~ 257 Co~ <NA>    Albe~ AL      35951 3927 No      Not Vi~
## 6 AL      Albertville~ 1100 H~ <NA>    Albe~ AL      35950 2532 No      Not Vi~
## 7 AL      Kate Duncan~ 6077 M~ <NA>    Grant AL      35747 <NA> No      Not Vi~
## 8 AL      Asbury High~ 1990 A~ <NA>    Albe~ AL      35951 <NA> No      Not Vi~
## 9 AL      Claysville ~ 140 Cl~ <NA>    Gunt~ AL      35976 <NA> No      Not Vi~
## 10 AL     Douglas Ele~ Highwa~ <NA>    Doug~ AL      35964 <NA> No      Not Vi~
## # i 100,561 more rows
## # i 52 more variables: SchoolLevel <chr>, SchoolType <chr>, Status <chr>,
## #   Locale <chr>, County <chr>, TotalFreeLunch <dbl>, FreeLunch <dbl>,
## #   ReducedLunch <dbl>, MealProgramCertified <dbl>, PreK <dbl>,
## #   Kindergarten <dbl>, Grade1 <dbl>, Grade2 <dbl>, Grade3 <dbl>, Grade4 <dbl>,
## #   Grade5 <dbl>, Grade6 <dbl>, Grade7 <dbl>, Grade8 <dbl>, Grade9 <dbl>,
```

```
## # Grade10 <dbl>, Grade11 <dbl>, Grade12 <dbl>, Grade13 <dbl>, ...
```

## Finding % of Students with Free/Reduced Lunch

```
df0 <- df0 %>%
  mutate(LunchRate = ((FreeLunch + ReducedLunch) / TotalEnrollment) * 100) %>%
  relocate(LunchRate, .after = MealProgramCertified)

df0
```

```
## # A tibble: 62,155 x 63
##   StateABR SchoolName Street1 Street2 City State Zip Zip4 Charter Virtual
##   <chr>      <chr>      <chr>  <chr>  <chr> <chr> <chr> <chr> <chr>  <chr>
## 1 AL      Albertville~ 600 E ~ <NA>  Albe~ AL   35950 <NA> No      Not Vi~
## 2 AL      Albertville~ 402 E ~ <NA>  Albe~ AL   35950 2322 No      Not Vi~
## 3 AL      Albertville~ 901 W ~ <NA>  Albe~ AL   35950 1300 No      Not Vi~
## 4 AL      Albertville~ 145 We~ <NA>  Albe~ AL   35950 <NA> No      Not Vi~
## 5 AL      Albertville~ 257 Co~ <NA>  Albe~ AL   35951 3927 No      Not Vi~
## 6 AL      Albertville~ 1100 H~ <NA>  Albe~ AL   35950 2532 No      Not Vi~
## 7 AL      Kate Duncan~ 6077 M~ <NA>  Grant AL   35747 <NA> No      Not Vi~
## 8 AL      Asbury High~ 1990 A~ <NA>  Albe~ AL   35951 <NA> No      Not Vi~
## 9 AL      Claysville ~ 140 Cl~ <NA>  Gunt~ AL   35976 <NA> No      Not Vi~
## 10 AL     Douglas Ele~ Highwa~ <NA>  Doug~ AL   35964 <NA> No      Not Vi~
## # i 62,145 more rows
## # i 53 more variables: SchoolLevel <chr>, SchoolType <chr>, Status <chr>,
## #   Locale <chr>, County <chr>, TotalFreeLunch <dbl>, FreeLunch <dbl>,
## #   ReducedLunch <dbl>, MealProgramCertified <dbl>, LunchRate <dbl>,
## #   PreK <dbl>, Kindergarten <dbl>, Grade1 <dbl>, Grade2 <dbl>, Grade3 <dbl>,
## #   Grade4 <dbl>, Grade5 <dbl>, Grade6 <dbl>, Grade7 <dbl>, Grade8 <dbl>,
## #   Grade9 <dbl>, Grade10 <dbl>, Grade11 <dbl>, Grade12 <dbl>, ...
```

```
df_all <- df_all %>%
  mutate(LunchRate = ((FreeLunch + ReducedLunch) / TotalEnrollment) * 100) %>%
  relocate(LunchRate, .after = MealProgramCertified)

df_all
```

```
## # A tibble: 100,571 x 63
##   StateABR SchoolName Street1 Street2 City State Zip Zip4 Charter Virtual
##   <chr>      <chr>      <chr>  <chr>  <chr> <chr> <chr> <chr> <chr>  <chr>
## 1 AL      Albertville~ 600 E ~ <NA>  Albe~ AL   35950 <NA> No      Not Vi~
## 2 AL      Albertville~ 402 E ~ <NA>  Albe~ AL   35950 2322 No      Not Vi~
## 3 AL      Albertville~ 901 W ~ <NA>  Albe~ AL   35950 1300 No      Not Vi~
## 4 AL      Albertville~ 145 We~ <NA>  Albe~ AL   35950 <NA> No      Not Vi~
## 5 AL      Albertville~ 257 Co~ <NA>  Albe~ AL   35951 3927 No      Not Vi~
## 6 AL      Albertville~ 1100 H~ <NA>  Albe~ AL   35950 2532 No      Not Vi~
## 7 AL      Kate Duncan~ 6077 M~ <NA>  Grant AL   35747 <NA> No      Not Vi~
## 8 AL      Asbury High~ 1990 A~ <NA>  Albe~ AL   35951 <NA> No      Not Vi~
## 9 AL      Claysville ~ 140 Cl~ <NA>  Gunt~ AL   35976 <NA> No      Not Vi~
## 10 AL     Douglas Ele~ Highwa~ <NA>  Doug~ AL   35964 <NA> No      Not Vi~
## # i 100,561 more rows
```

```
## # i 53 more variables: SchoolLevel <chr>, SchoolType <chr>, Status <chr>,
## #   Locale <chr>, County <chr>, TotalFreeLunch <dbl>, FreeLunch <dbl>,
## #   ReducedLunch <dbl>, MealProgramCertified <dbl>, LunchRate <dbl>,
## #   PreK <dbl>, Kindergarten <dbl>, Grade1 <dbl>, Grade2 <dbl>, Grade3 <dbl>,
## #   Grade4 <dbl>, Grade5 <dbl>, Grade6 <dbl>, Grade7 <dbl>, Grade8 <dbl>,
## #   Grade9 <dbl>, Grade10 <dbl>, Grade11 <dbl>, Grade12 <dbl>, ...
```

## Get rid of Incorrect Values for Key Variables

```
df0 <- df0 %>% filter(FreeLunch >= 0) %>% filter(ReducedLunch >= 0) %>% filter(MealProgramCertified >= 0)
df0
```

```
## # A tibble: 30,900 x 63
##   StateABR SchoolName Street1 Street2 City State Zip Zip4 Charter Virtual
##   <chr>      <chr>      <chr>  <chr>  <chr> <chr> <chr> <chr> <chr>  <chr>
## 1 AL      Albertville~ 600 E ~ <NA>  Albe~ AL   35950 <NA> No      Not Vi~
## 2 AL      Albertville~ 402 E ~ <NA>  Albe~ AL   35950 2322 No      Not Vi~
## 3 AL      Albertville~ 901 W ~ <NA>  Albe~ AL   35950 1300 No      Not Vi~
## 4 AL      Albertville~ 145 We~ <NA>  Albe~ AL   35950 <NA> No      Not Vi~
## 5 AL      Albertville~ 257 Co~ <NA>  Albe~ AL   35951 3927 No      Not Vi~
## 6 AL      Albertville~ 1100 H~ <NA>  Albe~ AL   35950 2532 No      Not Vi~
## 7 AL      Kate Duncan~ 6077 M~ <NA>  Grant AL   35747 <NA> No      Not Vi~
## 8 AL      Asbury High~ 1990 A~ <NA>  Albe~ AL   35951 <NA> No      Not Vi~
## 9 AL      Douglas Ele~ Highwa~ <NA>  Doug~ AL   35964 <NA> No      Not Vi~
## 10 AL     Douglas Hig~ 225 Ea~ <NA>  Doug~ AL   35964 <NA> No      Not Vi~
## # i 30,890 more rows
## # i 53 more variables: SchoolLevel <chr>, SchoolType <chr>, Status <chr>,
## #   Locale <chr>, County <chr>, TotalFreeLunch <dbl>, FreeLunch <dbl>,
## #   ReducedLunch <dbl>, MealProgramCertified <dbl>, LunchRate <dbl>,
## #   PreK <dbl>, Kindergarten <dbl>, Grade1 <dbl>, Grade2 <dbl>, Grade3 <dbl>,
## #   Grade4 <dbl>, Grade5 <dbl>, Grade6 <dbl>, Grade7 <dbl>, Grade8 <dbl>,
## #   Grade9 <dbl>, Grade10 <dbl>, Grade11 <dbl>, Grade12 <dbl>, ...
```

```
df_all <- df_all %>% filter(FreeLunch >= 0) %>% filter(ReducedLunch >= 0) %>% filter(MealProgramCertified >= 0)
df_all
```

```
## # A tibble: 39,775 x 63
##   StateABR SchoolName Street1 Street2 City State Zip Zip4 Charter Virtual
##   <chr>      <chr>      <chr>  <chr>  <chr> <chr> <chr> <chr> <chr>  <chr>
## 1 AL      Albertville~ 600 E ~ <NA>  Albe~ AL   35950 <NA> No      Not Vi~
## 2 AL      Albertville~ 402 E ~ <NA>  Albe~ AL   35950 2322 No      Not Vi~
## 3 AL      Albertville~ 901 W ~ <NA>  Albe~ AL   35950 1300 No      Not Vi~
## 4 AL      Albertville~ 145 We~ <NA>  Albe~ AL   35950 <NA> No      Not Vi~
## 5 AL      Albertville~ 257 Co~ <NA>  Albe~ AL   35951 3927 No      Not Vi~
## 6 AL      Albertville~ 1100 H~ <NA>  Albe~ AL   35950 2532 No      Not Vi~
## 7 AL      Kate Duncan~ 6077 M~ <NA>  Grant AL   35747 <NA> No      Not Vi~
## 8 AL      Asbury High~ 1990 A~ <NA>  Albe~ AL   35951 <NA> No      Not Vi~
## 9 AL      Douglas Ele~ Highwa~ <NA>  Doug~ AL   35964 <NA> No      Not Vi~
## 10 AL     Douglas Hig~ 225 Ea~ <NA>  Doug~ AL   35964 <NA> No      Not Vi~
## # i 39,765 more rows
## # i 53 more variables: SchoolLevel <chr>, SchoolType <chr>, Status <chr>,
```



```
## #   Locale <chr>, County <chr>, TotalFreeLunch <dbl>, FreeLunch <dbl>,
## #   ReducedLunch <dbl>, MealProgramCertified <dbl>, LunchRate <dbl>,
## #   PreK <dbl>, Kindergarten <dbl>, Grade1 <dbl>, Grade2 <dbl>, Grade3 <dbl>,
## #   Grade4 <dbl>, Grade5 <dbl>, Grade6 <dbl>, Grade7 <dbl>, Grade8 <dbl>,
## #   Grade9 <dbl>, Grade10 <dbl>, Grade11 <dbl>, Grade12 <dbl>, ...
```

## Exploring the Data

### Summary of Each Dataframe

```
str(df0) #structure
```

```
## tibble [30,900 x 63] (S3: tbl_df/tbl/data.frame)
## $ StateABR           : chr [1:30900] "AL" "AL" "AL" "AL" ...
## $ SchoolName         : chr [1:30900] "Albertville Middle School" "Albertville High School" "Albert
## $ Street1            : chr [1:30900] "600 E Alabama Ave" "402 E McCord Ave" "901 W McKinney Ave" "
## $ Street2            : chr [1:30900] NA NA NA NA ...
## $ City               : chr [1:30900] "Albertville" "Albertville" "Albertville" "Albertville" ...
## $ State              : chr [1:30900] "AL" "AL" "AL" "AL" ...
## $ Zip                : chr [1:30900] "35950" "35950" "35950" "35950" ...
## $ Zip4               : chr [1:30900] NA "2322" "1300" NA ...
## $ Charter            : chr [1:30900] "No" "No" "No" "No" ...
## $ Virtual            : chr [1:30900] "Not Virtual" "Not Virtual" "Not Virtual" "Not Virtual" ...
## $ SchoolLevel        : chr [1:30900] "Middle" "High" "Middle" "Elementary" ...
## $ SchoolType         : chr [1:30900] "Regular School" "Regular School" "Regular School" "Regular S
## $ Status             : chr [1:30900] "Currently operational" "Currently operational" "Currently op
## $ Locale             : chr [1:30900] "32-Town: Distant" "32-Town: Distant" "32-Town: Distant" "32-
## $ County             : chr [1:30900] "Marshall County" "Marshall County" "Marshall County" "Marsha
## $ TotalFreeLunch     : num [1:30900] 697 1254 718 723 392 ...
## $ FreeLunch          : num [1:30900] 654 1178 665 680 367 ...
## $ ReducedLunch       : num [1:30900] 43 76 53 43 25 53 28 54 22 14 ...
## $ MealProgramCertified: num [1:30900] 587 1059 570 583 240 ...
## $ LunchRate          : num [1:30900] 78.3 73.2 82.2 82.7 64.7 ...
## $ PreK               : num [1:30900] NA NA NA NA 133 0 NA NA 0 NA ...
## $ Kindergarten       : num [1:30900] NA NA NA NA 473 NA NA NA NA NA ...
## $ Grade1             : num [1:30900] NA NA NA NA NA 427 NA NA NA NA ...
## $ Grade2             : num [1:30900] NA NA NA NA NA 517 NA NA NA NA ...
## $ Grade3             : num [1:30900] NA NA NA 430 NA NA NA NA 141 NA ...
## $ Grade4             : num [1:30900] NA NA NA 444 NA NA NA NA 158 NA ...
## $ Grade5             : num [1:30900] NA NA 412 NA NA NA 95 NA 162 NA ...
## $ Grade6             : num [1:30900] NA NA 462 NA NA NA 97 104 NA NA ...
## $ Grade7             : num [1:30900] 440 NA NA NA NA NA 86 91 NA NA ...
## $ Grade8             : num [1:30900] 450 NA NA NA NA NA 86 108 NA NA ...
## $ Grade9             : num [1:30900] NA 493 NA NA NA NA NA 103 NA 202 ...
## $ Grade10            : num [1:30900] NA 442 NA NA NA NA NA 86 NA 177 ...
## $ Grade11            : num [1:30900] NA 390 NA NA NA NA NA 87 NA 139 ...
## $ Grade12            : num [1:30900] NA 387 NA NA NA NA NA 85 NA 156 ...
## $ Grade13            : num [1:30900] NA NA NA NA NA NA NA NA NA NA ...
## $ Ungrade            : num [1:30900] NA NA NA NA NA NA NA NA NA NA ...
## $ MaleEnrollment     : num [1:30900] 459 868 451 463 304 498 192 350 222 351 ...
## $ FemEnrollment      : num [1:30900] 431 844 423 411 302 446 172 314 239 323 ...
## $ TotalEnrollment    : num [1:30900] 890 1712 874 874 606 ...
```



```
## $ StudentTeacherRatio : num [1:30900] 19.8 20.1 20.3 20.3 23.3 ...
## $ AIANMale             : num [1:30900] 4 0 1 0 1 9 1 1 3 1 ...
## $ AIANFemale           : num [1:30900] 1 2 4 4 3 1 3 1 3 3 ...
## $ AIANTotal            : num [1:30900] 5 2 5 4 4 10 4 2 6 4 ...
## $ AsianMale            : num [1:30900] 4 4 4 1 2 3 0 1 0 0 ...
## $ AsianFemale          : num [1:30900] 2 5 0 3 0 0 0 1 0 0 ...
## $ AsianTotal           : num [1:30900] 6 9 4 4 2 3 0 2 0 0 ...
## $ BlackMale            : num [1:30900] 15 23 22 22 26 24 2 8 4 6 ...
## $ BlackFemale          : num [1:30900] 14 34 28 16 23 21 0 3 5 4 ...
## $ BlackTotal           : num [1:30900] 29 57 50 38 49 45 2 11 9 10 ...
## $ HPIMale              : num [1:30900] 0 0 0 0 0 0 0 0 0 2 ...
## $ HPIFemale            : num [1:30900] 1 0 0 0 0 1 0 2 1 0 ...
## $ HPITotal             : num [1:30900] 1 0 0 0 0 1 0 2 1 2 ...
## $ HispanicMale         : num [1:30900] 251 490 263 261 167 290 6 162 92 153 ...
## $ HispanicFemale       : num [1:30900] 251 468 241 236 152 256 8 156 102 140 ...
## $ HispanicTotal        : num [1:30900] 502 958 504 497 319 546 14 318 194 293 ...
## $ TRMale               : num [1:30900] 17 26 7 11 4 9 5 2 2 1 ...
## $ TRFemale             : num [1:30900] 15 19 6 16 4 10 9 3 3 2 ...
## $ TRTotal              : num [1:30900] 32 45 13 27 8 19 14 5 5 3 ...
## $ WhiteMale            : num [1:30900] 168 325 154 168 104 163 178 176 121 188 ...
## $ WhiteFemale          : num [1:30900] 147 316 144 136 120 157 152 148 125 174 ...
## $ WhiteTotal           : num [1:30900] 315 641 298 304 224 320 330 324 246 362 ...
## $ Latitude             : num [1:30900] 34.3 34.3 34.3 34.3 34.3 ...
## $ Longitude            : num [1:30900] -86.2 -86.2 -86.2 -86.2 -86.2 ...
```

```
dim(df0) #dimensions
```

```
## [1] 30900    63
```

```
summary (df0)
```

```
##      StateABR      SchoolName      Street1      Street2
## Length:30900    Length:30900    Length:30900    Length:30900
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##      City      State      Zip      Zip4
## Length:30900    Length:30900    Length:30900    Length:30900
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##      Charter      Virtual      SchoolLevel      SchoolType
## Length:30900    Length:30900    Length:30900    Length:30900
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
```

```

##
##
##      Status      Locale      County      TotalFreeLunch
## Length:30900      Length:30900      Length:30900      Min.   :   3.0
## Class :character   Class :character   Class :character   1st Qu.: 135.0
## Mode  :character   Mode  :character   Mode  :character   Median : 260.0
##                                     Mean  : 323.5
##                                     3rd Qu.: 422.0
##                                     Max.   :3979.0
##
##      FreeLunch      ReducedLunch      MealProgramCertified      LunchRate
## Min.   :   0      Min.   :   0.00      Min.   :   3.0      Min.   :   0.3836
## 1st Qu.: 115      1st Qu.:   7.00      1st Qu.:  73.0      1st Qu.: 38.4615
## Median : 227      Median :  22.00      Median : 155.0      Median : 60.2837
## Mean   : 287      Mean   :  36.53      Mean   : 205.1      Mean   : 59.4754
## 3rd Qu.: 379      3rd Qu.:  46.00      3rd Qu.: 277.0      3rd Qu.: 81.8966
## Max.   :3348      Max.   :1367.00      Max.   :2697.0      Max.   :100.0000
##
##      PreK      Kindergarten      Grade1      Grade2
## Min.   :   0.0      Min.   :   0.00      Min.   :   0.00      Min.   :   0.00
## 1st Qu.: 10.0      1st Qu.: 46.00      1st Qu.: 45.00      1st Qu.: 44.00
## Median : 24.0      Median : 70.00      Median : 68.00      Median : 67.00
## Mean   : 32.8      Mean   : 73.53      Mean   : 71.61      Mean   : 69.93
## 3rd Qu.: 43.0      3rd Qu.: 96.00      3rd Qu.: 93.00      3rd Qu.: 90.00
## Max.   :903.0      Max.   :873.00      Max.   :638.00      Max.   :624.00
## NA's   :20969      NA's   :11327      NA's   :11273      NA's   :11260
##      Grade3      Grade4      Grade5      Grade6
## Min.   :   0.00      Min.   :   0.00      Min.   :   0.00      Min.   :   0.0
## 1st Qu.: 46.00      1st Qu.: 45.00      1st Qu.: 45.00      1st Qu.: 37.0
## Median : 69.00      Median : 68.00      Median : 68.00      Median : 72.0
## Mean   : 72.27      Mean   : 71.46      Mean   : 73.07      Mean   :108.8
## 3rd Qu.: 94.00      3rd Qu.: 92.00      3rd Qu.: 94.00      3rd Qu.:144.0
## Max.   :597.00      Max.   :645.00      Max.   :727.00      Max.   :923.0
## NA's   :11310      NA's   :11385      NA's   :11711      NA's   :18326
##      Grade7      Grade8      Grade9      Grade10
## Min.   :   0.0      Min.   :   0.0      Min.   :   0.0      Min.   :   0.0
## 1st Qu.: 37.0      1st Qu.: 36.0      1st Qu.: 49.0      1st Qu.: 48.0
## Median : 97.0      Median : 98.0      Median : 137.0      Median : 134.0
## Mean   :142.8      Mean   :145.2      Mean   : 227.8      Mean   : 221.1
## 3rd Qu.:229.0      3rd Qu.:233.0      3rd Qu.: 378.0      3rd Qu.: 364.0
## Max.   :844.0      Max.   :875.0      Max.   :1307.0      Max.   :2855.0
## NA's   :21443      NA's   :21469      NA's   :25261      NA's   :25329
##      Grade11      Grade12      Grade13      Ungrade
## Min.   :   0.0      Min.   :   0.0      Min.   : NA      Min.   :   0.000
## 1st Qu.: 44.0      1st Qu.: 42.0      1st Qu.: NA      1st Qu.:   0.000
## Median : 122.0      Median : 115.0      Median : NA      Median :   3.000
## Mean   : 204.4      Mean   : 192.8      Mean   :NaN      Mean   :   5.991
## 3rd Qu.: 333.0      3rd Qu.: 310.0      3rd Qu.: NA      3rd Qu.:   9.000
## Max.   :1293.0      Max.   :1339.0      Max.   : NA      Max.   :223.000
## NA's   :25346      NA's   :25358      NA's   :30900      NA's   :29203
## MaleEnrollment      FemEnrollment      TotalEnrollment      StudentTeacherRatio
## Min.   :   0.0      Min.   :   0      Min.   :   9.0      Min.   :   1.07
## 1st Qu.: 159.0      1st Qu.: 150      1st Qu.: 310.0      1st Qu.: 13.76
## Median : 243.0      Median : 229      Median : 472.0      Median : 16.50

```

|    |              |          |                |          |               |           |             |          |
|----|--------------|----------|----------------|----------|---------------|-----------|-------------|----------|
| ## | Mean         | : 292.1  | Mean           | : 277    | Mean          | : 568.8   | Mean        | : 17.47  |
| ## | 3rd Qu.:     | 352.0    | 3rd Qu.:       | 333      | 3rd Qu.:      | 682.0     | 3rd Qu.:    | 20.37    |
| ## | Max.         | :3521.0  | Max.           | :3379    | Max.          | :6886.0   | Max.        | :1860.00 |
| ## | NA's         | :629     | NA's           | :629     |               |           |             |          |
| ## | AIANMale     |          | AIANFemale     |          | AIANTotal     |           | AsianMale   |          |
| ## | Min.         | : 0.000  | Min.           | : 0.000  | Min.          | : 0.000   | Min.        | : 0.00   |
| ## | 1st Qu.:     | 0.000    | 1st Qu.:       | 0.000    | 1st Qu.:      | 0.000     | 1st Qu.:    | 0.00     |
| ## | Median       | : 0.000  | Median         | : 0.000  | Median        | : 1.000   | Median      | : 3.00   |
| ## | Mean         | : 3.587  | Mean           | : 3.474  | Mean          | : 7.057   | Mean        | : 17.62  |
| ## | 3rd Qu.:     | 2.000    | 3rd Qu.:       | 2.000    | 3rd Qu.:      | 3.000     | 3rd Qu.:    | 13.00    |
| ## | Max.         | :585.000 | Max.           | :513.000 | Max.          | :1098.000 | Max.        | :1335.00 |
| ## | NA's         | :666     | NA's           | :670     | NA's          | :648      | NA's        | :632     |
| ## | AsianFemale  |          | AsianTotal     |          | BlackMale     |           | BlackFemale |          |
| ## | Min.         | : 0.00   | Min.           | : 0.00   | Min.          | : 0.00    | Min.        | : 0.00   |
| ## | 1st Qu.:     | 0.00     | 1st Qu.:       | 1.00     | 1st Qu.:      | 1.00      | 1st Qu.:    | 1.00     |
| ## | Median       | : 3.00   | Median         | : 6.00   | Median        | : 10.00   | Median      | : 9.00   |
| ## | Mean         | : 16.49  | Mean           | : 34.11  | Mean          | : 43.69   | Mean        | : 42.21  |
| ## | 3rd Qu.:     | 12.00    | 3rd Qu.:       | 26.00    | 3rd Qu.:      | 50.00     | 3rd Qu.:    | 48.00    |
| ## | Max.         | :1224.00 | Max.           | :2559.00 | Max.          | :1075.00  | Max.        | :1220.00 |
| ## | NA's         | :631     | NA's           | :629     | NA's          | :635      | NA's        | :638     |
| ## | BlackTotal   |          | HPIMale        |          | HPIFemale     |           | HPITotal    |          |
| ## | Min.         | : 0.00   | Min.           | : 0.000  | Min.          | : 0.000   | Min.        | : 0.000  |
| ## | 1st Qu.:     | 3.00     | 1st Qu.:       | 0.000    | 1st Qu.:      | 0.000     | 1st Qu.:    | 0.000    |
| ## | Median       | : 19.00  | Median         | : 0.000  | Median        | : 0.000   | Median      | : 0.000  |
| ## | Mean         | : 85.89  | Mean           | : 1.869  | Mean          | : 1.761   | Mean        | : 3.627  |
| ## | 3rd Qu.:     | 98.00    | 3rd Qu.:       | 1.000    | 3rd Qu.:      | 1.000     | 3rd Qu.:    | 2.000    |
| ## | Max.         | :2295.00 | Max.           | :556.000 | Max.          | :440.000  | Max.        | :996.000 |
| ## | NA's         | :633     | NA's           | :671     | NA's          | :671      | NA's        | :653     |
| ## | HispanicMale |          | HispanicFemale |          | HispanicTotal |           | TRMale      |          |
| ## | Min.         | : 0.00   | Min.           | : 0.00   | Min.          | : 0.0     | Min.        | : 0.00   |
| ## | 1st Qu.:     | 12.00    | 1st Qu.:       | 11.00    | 1st Qu.:      | 23.0      | 1st Qu.:    | 3.00     |
| ## | Median       | : 43.00  | Median         | : 41.00  | Median        | : 84.0    | Median      | : 10.00  |
| ## | Mean         | : 92.68  | Mean           | : 88.61  | Mean          | : 181.3   | Mean        | : 15.46  |
| ## | 3rd Qu.:     | 125.00   | 3rd Qu.:       | 119.00   | 3rd Qu.:      | 245.0     | 3rd Qu.:    | 22.00    |
| ## | Max.         | :1947.00 | Max.           | :2118.00 | Max.          | :4065.0   | Max.        | :376.00  |
| ## | NA's         | :629     | NA's           | :629     | NA's          | :629      | NA's        | :632     |
| ## | TRFemale     |          | TRTotal        |          | WhiteMale     |           | WhiteFemale |          |
| ## | Min.         | : 0.00   | Min.           | : 0.00   | Min.          | : 0.0     | Min.        | : 0.0    |
| ## | 1st Qu.:     | 3.00     | 1st Qu.:       | 7.00     | 1st Qu.:      | 23.0      | 1st Qu.:    | 21.0     |
| ## | Median       | : 10.00  | Median         | : 21.00  | Median        | : 85.0    | Median      | : 78.0   |
| ## | Mean         | : 14.83  | Mean           | : 30.29  | Mean          | : 117.2   | Mean        | : 109.7  |
| ## | 3rd Qu.:     | 21.00    | 3rd Qu.:       | 42.00    | 3rd Qu.:      | 167.0     | 3rd Qu.:    | 156.0    |
| ## | Max.         | :422.00  | Max.           | :798.00  | Max.          | :1775.0   | Max.        | :1761.0  |
| ## | NA's         | :631     | NA's           | :631     | NA's          | :629      | NA's        | :629     |
| ## | WhiteTotal   |          | Latitude       |          | Longitude     |           |             |          |
| ## | Min.         | : 0.0    | Min.           | :14.14   | Min.          | : -171.72 |             |          |
| ## | 1st Qu.:     | 45.0     | 1st Qu.:       | 33.51    | 1st Qu.:      | -118.26   |             |          |
| ## | Median       | : 164.0  | Median         | :36.29   | Median        | : -94.57  |             |          |
| ## | Mean         | : 226.9  | Mean           | :36.88   | Mean          | : -101.46 |             |          |
| ## | 3rd Qu.:     | 323.0    | 3rd Qu.:       | 39.79    | 3rd Qu.:      | -85.76    |             |          |
| ## | Max.         | :3536.0  | Max.           | :67.73   | Max.          | : 145.78  |             |          |
| ## | NA's         | :629     |                |          |               |           |             |          |

glimpse (df0)

```
## Rows: 30,900
## Columns: 63
## $ StateABR      <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", "~
## $ SchoolName    <chr> "Albertville Middle School", "Albertville High Sc~
## $ Street1       <chr> "600 E Alabama Ave", "402 E McCord Ave", "901 W M~
## $ Street2       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ City          <chr> "Albertville", "Albertville", "Albertville", "Alb~
## $ State         <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", "~
## $ Zip           <chr> "35950", "35950", "35950", "35950", "35951", "359~
## $ Zip4          <chr> NA, "2322", "1300", NA, "3927", "2532", NA, NA, N~
## $ Charter       <chr> "No", "No", "No", "No", "No", "No", "No", "No", "~
## $ Virtual       <chr> "Not Virtual", "Not Virtual", "Not Virtual", "Not~
## $ SchoolLevel   <chr> "Middle", "High", "Middle", "Elementary", "Elemen~
## $ SchoolType    <chr> "Regular School", "Regular School", "Regular Scho~
## $ Status        <chr> "Currently operational", "Currently operational",~
## $ Locale        <chr> "32-Town: Distant", "32-Town: Distant", "32-Town:~
## $ County        <chr> "Marshall County", "Marshall County", "Marshall C~
## $ TotalFreeLunch <dbl> 697, 1254, 718, 723, 392, 779, 151, 555, 245, 334~
## $ FreeLunch     <dbl> 654, 1178, 665, 680, 367, 726, 123, 501, 223, 320~
## $ ReducedLunch  <dbl> 43, 76, 53, 43, 25, 53, 28, 54, 22, 14, 10, 17, 1~
## $ MealProgramCertified <dbl> 587, 1059, 570, 583, 240, 617, 194, 491, 316, 438~
## $ LunchRate     <dbl> 78.31461, 73.24766, 82.15103, 82.72311, 64.68647,~
## $ PreK          <dbl> NA, NA, NA, NA, 133, 0, NA, NA, 0, NA, 38, NA, 0,~
## $ Kindergarten  <dbl> NA, NA, NA, NA, 473, NA, NA, NA, NA, NA, NA, NA, ~
## $ Grade1        <dbl> NA, NA, NA, NA, NA, 427, NA, NA, NA, NA, NA, NA, ~
## $ Grade2        <dbl> NA, NA, NA, NA, NA, 517, NA, NA, NA, NA, NA, NA, ~
## $ Grade3        <dbl> NA, NA, NA, 430, NA, NA, NA, NA, 141, NA, 77, NA,~
## $ Grade4        <dbl> NA, NA, NA, 444, NA, NA, NA, NA, 158, NA, 52, NA,~
## $ Grade5        <dbl> NA, NA, 412, NA, NA, NA, 95, NA, 162, NA, 69, NA,~
## $ Grade6        <dbl> NA, NA, 462, NA, NA, NA, 97, 104, NA, NA, NA, NA,~
## $ Grade7        <dbl> 440, NA, NA, NA, NA, NA, 86, 91, NA, NA, NA, NA, ~
## $ Grade8        <dbl> 450, NA, NA, NA, NA, NA, 86, 108, NA, NA, NA, NA,~
## $ Grade9        <dbl> NA, 493, NA, NA, NA, NA, NA, 103, NA, 202, NA, 10~
## $ Grade10       <dbl> NA, 442, NA, NA, NA, NA, NA, 86, NA, 177, NA, 101~
## $ Grade11       <dbl> NA, 390, NA, NA, NA, NA, NA, 87, NA, 139, NA, 96,~
## $ Grade12       <dbl> NA, 387, NA, NA, NA, NA, NA, 85, NA, 156, NA, 92,~
## $ Grade13       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ Ungrade       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ MaleEnrollment <dbl> 459, 868, 451, 463, 304, 498, 192, 350, 222, 351,~
## $ FemEnrollment <dbl> 431, 844, 423, 411, 302, 446, 172, 314, 239, 323,~
## $ TotalEnrollment <dbl> 890, 1712, 874, 874, 606, 944, 364, 664, 461, 674~
## $ StudentTeacherRatio <dbl> 19.78, 20.09, 20.33, 20.33, 23.31, 15.48, 16.52, ~
## $ AIANMale      <dbl> 4, 0, 1, 0, 1, 9, 1, 1, 3, 1, 0, 1, 0, 1, 3, 4, 0~
## $ AIANFemale    <dbl> 1, 2, 4, 4, 3, 1, 3, 1, 3, 3, 0, 1, 0, 0, 1, 2, 2~
## $ AIANTotal     <dbl> 5, 2, 5, 4, 4, 10, 4, 2, 6, 4, 0, 2, 0, 1, 4, 6, ~
## $ AsianMale     <dbl> 4, 4, 4, 1, 2, 3, 0, 1, 0, 0, 4, 1, 0, 0, 2, 0, 1~
## $ AsianFemale   <dbl> 2, 5, 0, 3, 0, 0, 0, 1, 0, 0, 0, 4, 0, 0, 3, 3, 1~
## $ AsianTotal    <dbl> 6, 9, 4, 4, 2, 3, 0, 2, 0, 0, 4, 5, 0, 0, 5, 3, 2~
## $ BlackMale     <dbl> 15, 23, 22, 22, 26, 24, 2, 8, 4, 6, 2, 3, 0, 7, 7~
## $ BlackFemale   <dbl> 14, 34, 28, 16, 23, 21, 0, 3, 5, 4, 1, 2, 0, 5, 3~
## $ BlackTotal    <dbl> 29, 57, 50, 38, 49, 45, 2, 11, 9, 10, 3, 5, 0, 12~
```

```
## $ HPIMale <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0~
## $ HPIFemale <dbl> 1, 0, 0, 0, 0, 1, 0, 2, 1, 0, 0, 0, 0, 0, 0, 0, 0~
## $ HPITotal <dbl> 1, 0, 0, 0, 0, 1, 0, 2, 1, 2, 0, 0, 0, 0, 0, 0, 0~
## $ HispanicMale <dbl> 251, 490, 263, 261, 167, 290, 6, 162, 92, 153, 2,~
## $ HispanicFemale <dbl> 251, 468, 241, 236, 152, 256, 8, 156, 102, 140, 5~
## $ HispanicTotal <dbl> 502, 958, 504, 497, 319, 546, 14, 318, 194, 293, ~
## $ TRMale <dbl> 17, 26, 7, 11, 4, 9, 5, 2, 2, 1, 0, 4, 4, 1, 4, 3~
## $ TRFemale <dbl> 15, 19, 6, 16, 4, 10, 9, 3, 3, 2, 3, 4, 3, 3, 4, ~
## $ TRTotal <dbl> 32, 45, 13, 27, 8, 19, 14, 5, 5, 3, 3, 8, 7, 4, 8~
## $ WhiteMale <dbl> 168, 325, 154, 168, 104, 163, 178, 176, 121, 188,~
## $ WhiteFemale <dbl> 147, 316, 144, 136, 120, 157, 152, 148, 125, 174,~
## $ WhiteTotal <dbl> 315, 641, 298, 304, 224, 320, 330, 324, 246, 362,~
## $ Latitude <dbl> 34.26020, 34.26220, 34.27330, 34.25270, 34.28980,~
## $ Longitude <dbl> -86.20620, -86.20490, -86.22010, -86.22181, -86.1~
```

```
str(df_all)
```

```
## tibble [39,775 x 63] (S3: tbl_df/tbl/data.frame)
## $ StateABR : chr [1:39775] "AL" "AL" "AL" "AL" ...
## $ SchoolName : chr [1:39775] "Albertville Middle School" "Albertville High School" "Albertville High School" ...
## $ Street1 : chr [1:39775] "600 E Alabama Ave" "402 E McCord Ave" "901 W McKinney Ave" ...
## $ Street2 : chr [1:39775] NA NA NA NA ...
## $ City : chr [1:39775] "Albertville" "Albertville" "Albertville" "Albertville" ...
## $ State : chr [1:39775] "AL" "AL" "AL" "AL" ...
## $ Zip : chr [1:39775] "35950" "35950" "35950" "35950" ...
## $ Zip4 : chr [1:39775] NA "2322" "1300" NA ...
## $ Charter : chr [1:39775] "No" "No" "No" "No" ...
## $ Virtual : chr [1:39775] "Not Virtual" "Not Virtual" "Not Virtual" "Not Virtual" ...
## $ SchoolLevel : chr [1:39775] "Middle" "High" "Middle" "Elementary" ...
## $ SchoolType : chr [1:39775] "Regular School" "Regular School" "Regular School" "Regular School" ...
## $ Status : chr [1:39775] "Currently operational" "Currently operational" "Currently operational" "Currently operational" ...
## $ Locale : chr [1:39775] "32-Town: Distant" "32-Town: Distant" "32-Town: Distant" "32-Town: Distant" ...
## $ County : chr [1:39775] "Marshall County" "Marshall County" "Marshall County" "Marshall County" ...
## $ TotalFreeLunch : num [1:39775] 697 1254 718 723 392 ...
## $ FreeLunch : num [1:39775] 654 1178 665 680 367 ...
## $ ReducedLunch : num [1:39775] 43 76 53 43 25 53 28 54 22 14 ...
## $ MealProgramCertified : num [1:39775] 587 1059 570 583 240 ...
## $ LunchRate : num [1:39775] 78.3 73.2 82.2 82.7 64.7 ...
## $ PreK : num [1:39775] NA NA NA NA 133 0 NA NA 0 NA ...
## $ Kindergarten : num [1:39775] NA NA NA NA 473 NA NA NA NA ...
## $ Grade1 : num [1:39775] NA NA NA NA 427 NA NA NA NA ...
## $ Grade2 : num [1:39775] NA NA NA NA NA 517 NA NA NA NA ...
## $ Grade3 : num [1:39775] NA NA NA 430 NA NA NA NA 141 NA ...
## $ Grade4 : num [1:39775] NA NA NA 444 NA NA NA NA 158 NA ...
## $ Grade5 : num [1:39775] NA NA 412 NA NA NA 95 NA 162 NA ...
## $ Grade6 : num [1:39775] NA NA 462 NA NA NA 97 104 NA NA ...
## $ Grade7 : num [1:39775] 440 NA NA NA NA NA 86 91 NA NA ...
## $ Grade8 : num [1:39775] 450 NA NA NA NA NA 86 108 NA NA ...
## $ Grade9 : num [1:39775] NA 493 NA NA NA NA NA 103 NA 202 ...
## $ Grade10 : num [1:39775] NA 442 NA NA NA NA NA 86 NA 177 ...
## $ Grade11 : num [1:39775] NA 390 NA NA NA NA NA 87 NA 139 ...
## $ Grade12 : num [1:39775] NA 387 NA NA NA NA NA 85 NA 156 ...
## $ Grade13 : num [1:39775] NA NA NA NA NA NA NA NA NA NA ...
## $ Ungrade : num [1:39775] NA NA NA NA NA NA NA NA NA NA ...
```

```
## $ MaleEnrollment      : num [1:39775] 459 868 451 463 304 498 192 350 222 351 ...
## $ FemEnrollment       : num [1:39775] 431 844 423 411 302 446 172 314 239 323 ...
## $ TotalEnrollment     : num [1:39775] 890 1712 874 874 606 ...
## $ StudentTeacherRatio : num [1:39775] 19.8 20.1 20.3 20.3 23.3 ...
## $ AIANMale            : num [1:39775] 4 0 1 0 1 9 1 1 3 1 ...
## $ AIANFemale          : num [1:39775] 1 2 4 4 3 1 3 1 3 3 ...
## $ AIANTotal           : num [1:39775] 5 2 5 4 4 10 4 2 6 4 ...
## $ AsianMale           : num [1:39775] 4 4 4 1 2 3 0 1 0 0 ...
## $ AsianFemale         : num [1:39775] 2 5 0 3 0 0 0 1 0 0 ...
## $ AsianTotal          : num [1:39775] 6 9 4 4 2 3 0 2 0 0 ...
## $ BlackMale           : num [1:39775] 15 23 22 22 26 24 2 8 4 6 ...
## $ BlackFemale         : num [1:39775] 14 34 28 16 23 21 0 3 5 4 ...
## $ BlackTotal          : num [1:39775] 29 57 50 38 49 45 2 11 9 10 ...
## $ HPIMale             : num [1:39775] 0 0 0 0 0 0 0 0 0 2 ...
## $ HPIFemale           : num [1:39775] 1 0 0 0 0 1 0 2 1 0 ...
## $ HPITotal            : num [1:39775] 1 0 0 0 0 1 0 2 1 2 ...
## $ HispanicMale        : num [1:39775] 251 490 263 261 167 290 6 162 92 153 ...
## $ HispanicFemale      : num [1:39775] 251 468 241 236 152 256 8 156 102 140 ...
## $ HispanicTotal       : num [1:39775] 502 958 504 497 319 546 14 318 194 293 ...
## $ TRMale              : num [1:39775] 17 26 7 11 4 9 5 2 2 1 ...
## $ TRFemale            : num [1:39775] 15 19 6 16 4 10 9 3 3 2 ...
## $ TRTotal             : num [1:39775] 32 45 13 27 8 19 14 5 5 3 ...
## $ WhiteMale           : num [1:39775] 168 325 154 168 104 163 178 176 121 188 ...
## $ WhiteFemale         : num [1:39775] 147 316 144 136 120 157 152 148 125 174 ...
## $ WhiteTotal          : num [1:39775] 315 641 298 304 224 320 330 324 246 362 ...
## $ Latitude            : num [1:39775] 34.3 34.3 34.3 34.3 34.3 ...
## $ Longitude           : num [1:39775] -86.2 -86.2 -86.2 -86.2 -86.2 ...
```

```
dim(df_all)
```

```
## [1] 39775    63
```

```
summary (df_all)
```

```
##      StateABR      SchoolName      Street1      Street2
## Length:39775    Length:39775    Length:39775    Length:39775
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##      City      State      Zip      Zip4
## Length:39775    Length:39775    Length:39775    Length:39775
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##      Charter      Virtual      SchoolLevel      SchoolType
## Length:39775    Length:39775    Length:39775    Length:39775
## Class :character Class :character Class :character Class :character
```

```

## Mode :character Mode :character Mode :character Mode :character
##
##
##
##
## Status Locale County TotalFreeLunch
## Length:39775 Length:39775 Length:39775 Min. : 3.0
## Class :character Class :character Class :character 1st Qu.: 118.0
## Mode :character Mode :character Mode :character Median : 246.0
## Mean : 314.2
## 3rd Qu.: 414.0
## Max. :5770.0
##
## FreeLunch ReducedLunch MealProgramCertified LunchRate
## Min. : 0.0 Min. : 0.0 Min. : 3.0 Min. : 0.3836
## 1st Qu.: 99.0 1st Qu.: 4.0 1st Qu.: 64.0 1st Qu.: 38.6189
## Median : 216.0 Median : 19.0 Median : 148.0 Median : 60.9576
## Mean : 280.3 Mean : 33.9 Mean : 202.2 Mean : 59.6682
## 3rd Qu.: 374.0 3rd Qu.: 43.0 3rd Qu.: 275.0 3rd Qu.: 81.8966
## Max. :5563.0 Max. :1400.0 Max. :2921.0 Max. :100.0000
##
## PreK Kindergarten Grade1 Grade2
## Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.00
## 1st Qu.: 10.00 1st Qu.: 43.00 1st Qu.: 42.00 1st Qu.: 41.00
## Median : 24.00 Median : 67.00 Median : 67.00 Median : 65.00
## Mean : 33.34 Mean : 70.64 Mean : 69.21 Mean : 67.57
## 3rd Qu.: 43.00 3rd Qu.: 94.00 3rd Qu.: 91.00 3rd Qu.: 89.00
## Max. :1013.00 Max. :873.00 Max. :646.00 Max. :665.00
## NA's :27567 NA's :16522 NA's :16429 NA's :16407
## Grade3 Grade4 Grade5 Grade6
## Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.0
## 1st Qu.: 42.00 1st Qu.: 42.00 1st Qu.: 41.00 1st Qu.: 28.0
## Median : 67.00 Median : 66.00 Median : 66.00 Median : 68.0
## Mean : 69.69 Mean : 69.01 Mean : 70.53 Mean :103.7
## 3rd Qu.: 92.00 3rd Qu.: 91.00 3rd Qu.: 93.00 3rd Qu.:139.0
## Max. :691.00 Max. :669.00 Max. :727.00 Max. :923.0
## NA's :16430 NA's :16462 NA's :16701 NA's :23860
## Grade7 Grade8 Grade9 Grade10
## Min. : 0.0 Min. : 0.0 Min. : 0.0 Min. : 0.0
## 1st Qu.: 23.0 1st Qu.: 23.0 1st Qu.: 16.0 1st Qu.: 19.0
## Median : 79.0 Median : 79.0 Median : 83.0 Median : 81.0
## Mean :128.2 Mean :130.8 Mean : 182.3 Mean : 176.9
## 3rd Qu.:210.0 3rd Qu.:215.0 3rd Qu.: 310.0 3rd Qu.: 298.0
## Max. :844.0 Max. :930.0 Max. :6251.0 Max. :2855.0
## NA's :26821 NA's :26830 NA's :29601 NA's :29626
## Grade11 Grade12 Grade13 Ungrade
## Min. : 0.0 Min. : 0.0 Min. : NA Min. : 0.00
## 1st Qu.: 23.0 1st Qu.: 27.0 1st Qu.: NA 1st Qu.: 0.00
## Median : 77.0 Median : 82.0 Median : NA Median : 2.00
## Mean : 165.6 Mean : 161.7 Mean : NaN Mean : 5.57
## 3rd Qu.: 266.0 3rd Qu.: 252.0 3rd Qu.: NA 3rd Qu.: 8.00
## Max. :1293.0 Max. :1339.0 Max. : NA Max. :223.00
## NA's :29631 NA's :29590 NA's :39775 NA's :37777
## MaleEnrollment FemEnrollment TotalEnrollment StudentTeacherRatio

```



|    |              |          |                |          |               |           |             |          |
|----|--------------|----------|----------------|----------|---------------|-----------|-------------|----------|
| ## | Min.         | : 0.0    | Min.           | : 0.0    | Min.          | : 9.0     | Min.        | : 1.00   |
| ## | 1st Qu.:     | 141.0    | 1st Qu.:       | 132.0    | 1st Qu.:      | 274.0     | 1st Qu.:    | 13.28    |
| ## | Median :     | 235.0    | Median :       | 221.0    | Median :      | 456.0     | Median :    | 16.05    |
| ## | Mean :       | 285.5    | Mean :         | 270.1    | Mean :        | 555.6     | Mean :      | 17.14    |
| ## | 3rd Qu.:     | 348.0    | 3rd Qu.:       | 329.0    | 3rd Qu.:      | 675.0     | 3rd Qu.:    | 20.00    |
| ## | Max.         | :4352.0  | Max.           | :4524.0  | Max.          | :8876.0   | Max.        | :1860.00 |
| ## | NA's         | :784     | NA's           | :784     |               |           |             |          |
| ## | AIANMale     |          | AIANFemale     |          | AIANTotal     |           | AsianMale   |          |
| ## | Min.         | : 0.000  | Min.           | : 0.000  | Min.          | : 0.000   | Min.        | : 0.00   |
| ## | 1st Qu.:     | 0.000    | 1st Qu.:       | 0.000    | 1st Qu.:      | 0.000     | 1st Qu.:    | 0.00     |
| ## | Median :     | 0.000    | Median :       | 0.000    | Median :      | 1.000     | Median :    | 3.00     |
| ## | Mean :       | 3.305    | Mean :         | 3.183    | Mean :        | 6.483     | Mean :      | 18.01    |
| ## | 3rd Qu.:     | 2.000    | 3rd Qu.:       | 2.000    | 3rd Qu.:      | 3.000     | 3rd Qu.:    | 13.00    |
| ## | Max.         | :585.000 | Max.           | :513.000 | Max.          | :1098.000 | Max.        | :1335.00 |
| ## | NA's         | :839     | NA's           | :840     | NA's          | :807      | NA's        | :791     |
| ## | AsianFemale  |          | AsianTotal     |          | BlackMale     |           | BlackFemale |          |
| ## | Min.         | : 0.00   | Min.           | : 0.00   | Min.          | : 0.0     | Min.        | : 0.00   |
| ## | 1st Qu.:     | 0.00     | 1st Qu.:       | 1.00     | 1st Qu.:      | 1.0       | 1st Qu.:    | 1.00     |
| ## | Median :     | 3.00     | Median :       | 5.00     | Median :      | 10.0      | Median :    | 9.00     |
| ## | Mean :       | 16.88    | Mean :         | 34.88    | Mean :        | 43.3      | Mean :      | 41.67    |
| ## | 3rd Qu.:     | 12.00    | 3rd Qu.:       | 25.00    | 3rd Qu.:      | 50.0      | 3rd Qu.:    | 47.00    |
| ## | Max.         | :1224.00 | Max.           | :2559.00 | Max.          | :2195.0   | Max.        | :2207.00 |
| ## | NA's         | :787     | NA's           | :785     | NA's          | :795      | NA's        | :798     |
| ## | BlackTotal   |          | HPIMale        |          | HPIFemale     |           | HPITotal    |          |
| ## | Min.         | : 0.00   | Min.           | : 0.000  | Min.          | : 0.000   | Min.        | : 0.000  |
| ## | 1st Qu.:     | 3.00     | 1st Qu.:       | 0.000    | 1st Qu.:      | 0.000     | 1st Qu.:    | 0.000    |
| ## | Median :     | 19.00    | Median :       | 0.000    | Median :      | 0.000     | Median :    | 0.000    |
| ## | Mean :       | 84.96    | Mean :         | 1.703    | Mean :        | 1.608     | Mean :      | 3.308    |
| ## | 3rd Qu.:     | 97.00    | 3rd Qu.:       | 1.000    | 3rd Qu.:      | 1.000     | 3rd Qu.:    | 2.000    |
| ## | Max.         | :4402.00 | Max.           | :556.000 | Max.          | :440.000  | Max.        | :996.000 |
| ## | NA's         | :790     | NA's           | :852     | NA's          | :847      | NA's        | :819     |
| ## | HispanicMale |          | HispanicFemale |          | HispanicTotal |           | TRMale      |          |
| ## | Min.         | : 0.00   | Min.           | : 0.00   | Min.          | : 0.0     | Min.        | : 0.00   |
| ## | 1st Qu.:     | 10.00    | 1st Qu.:       | 9.00     | 1st Qu.:      | 20.0      | 1st Qu.:    | 3.00     |
| ## | Median :     | 37.00    | Median :       | 35.00    | Median :      | 72.0      | Median :    | 10.00    |
| ## | Mean :       | 87.14    | Mean :         | 83.02    | Mean :        | 170.2     | Mean :      | 15.52    |
| ## | 3rd Qu.:     | 113.00   | 3rd Qu.:       | 108.00   | 3rd Qu.:      | 221.0     | 3rd Qu.:    | 22.00    |
| ## | Max.         | :1947.00 | Max.           | :2118.00 | Max.          | :4065.0   | Max.        | :436.00  |
| ## | NA's         | :784     | NA's           | :784     | NA's          | :784      | NA's        | :787     |
| ## | TRFemale     |          | TRTotal        |          | WhiteMale     |           | WhiteFemale |          |
| ## | Min.         | : 0.00   | Min.           | : 0.00   | Min.          | : 0.0     | Min.        | : 0.0    |
| ## | 1st Qu.:     | 3.00     | 1st Qu.:       | 6.00     | 1st Qu.:      | 21.0      | 1st Qu.:    | 19.0     |
| ## | Median :     | 10.00    | Median :       | 20.00    | Median :      | 81.0      | Median :    | 74.0     |
| ## | Mean :       | 14.89    | Mean :         | 30.41    | Mean :        | 116.5     | Mean :      | 108.8    |
| ## | 3rd Qu.:     | 21.00    | 3rd Qu.:       | 42.00    | 3rd Qu.:      | 166.0     | 3rd Qu.:    | 154.0    |
| ## | Max.         | :422.00  | Max.           | :828.00  | Max.          | :1989.0   | Max.        | :2305.0  |
| ## | NA's         | :786     | NA's           | :786     | NA's          | :784      | NA's        | :784     |
| ## | WhiteTotal   |          | Latitude       |          | Longitude     |           |             |          |
| ## | Min.         | : 0.0    | Min.           | :14.14   | Min.          | : -171.72 |             |          |
| ## | 1st Qu.:     | 40.0     | 1st Qu.:       | 33.77    | 1st Qu.:      | -118.27   |             |          |
| ## | Median :     | 156.0    | Median :       | 37.01    | Median :      | -94.24    |             |          |
| ## | Mean :       | 225.3    | Mean :         | 37.34    | Mean :        | -100.71   |             |          |
| ## | 3rd Qu.:     | 320.0    | 3rd Qu.:       | 39.81    | 3rd Qu.:      | -84.58    |             |          |
| ## | Max.         | :4294.0  | Max.           | :71.30   | Max.          | : 145.78  |             |          |

## NA's :784

`glimpse` (df\_all)

```
## Rows: 39,775
## Columns: 63
## $ StateABR <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", "~
## $ SchoolName <chr> "Albertville Middle School", "Albertville High Sc~
## $ Street1 <chr> "600 E Alabama Ave", "402 E McCord Ave", "901 W M~
## $ Street2 <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ City <chr> "Albertville", "Albertville", "Albertville", "Alb~
## $ State <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", "~
## $ Zip <chr> "35950", "35950", "35950", "35950", "35951", "359~
## $ Zip4 <chr> NA, "2322", "1300", NA, "3927", "2532", NA, NA, N~
## $ Charter <chr> "No", "No", "No", "No", "No", "No", "No", "No", "~
## $ Virtual <chr> "Not Virtual", "Not Virtual", "Not Virtual", "Not~
## $ SchoolLevel <chr> "Middle", "High", "Middle", "Elementary", "Elemen~
## $ SchoolType <chr> "Regular School", "Regular School", "Regular Scho~
## $ Status <chr> "Currently operational", "Currently operational",~
## $ Locale <chr> "32-Town: Distant", "32-Town: Distant", "32-Town:~
## $ County <chr> "Marshall County", "Marshall County", "Marshall C~
## $ TotalFreeLunch <dbl> 697, 1254, 718, 723, 392, 779, 151, 555, 245, 334~
## $ FreeLunch <dbl> 654, 1178, 665, 680, 367, 726, 123, 501, 223, 320~
## $ ReducedLunch <dbl> 43, 76, 53, 43, 25, 53, 28, 54, 22, 14, 10, 17, 1~
## $ MealProgramCertified <dbl> 587, 1059, 570, 583, 240, 617, 194, 491, 316, 438~
## $ LunchRate <dbl> 78.31461, 73.24766, 82.15103, 82.72311, 64.68647,~
## $ PreK <dbl> NA, NA, NA, NA, 133, 0, NA, NA, 0, NA, 38, NA, 0,~
## $ Kindergarten <dbl> NA, NA, NA, NA, 473, NA, NA, NA, NA, NA, NA, NA,~
## $ Grade1 <dbl> NA, NA, NA, NA, NA, 427, NA, NA, NA, NA, NA, NA,~
## $ Grade2 <dbl> NA, NA, NA, NA, NA, 517, NA, NA, NA, NA, NA, NA,~
## $ Grade3 <dbl> NA, NA, NA, 430, NA, NA, NA, NA, 141, NA, 77, NA,~
## $ Grade4 <dbl> NA, NA, NA, 444, NA, NA, NA, NA, 158, NA, 52, NA,~
## $ Grade5 <dbl> NA, NA, 412, NA, NA, NA, 95, NA, 162, NA, 69, NA,~
## $ Grade6 <dbl> NA, NA, 462, NA, NA, NA, 97, 104, NA, NA, NA, NA,~
## $ Grade7 <dbl> 440, NA, NA, NA, NA, NA, 86, 91, NA, NA, NA, NA,~
## $ Grade8 <dbl> 450, NA, NA, NA, NA, NA, 86, 108, NA, NA, NA, NA,~
## $ Grade9 <dbl> NA, 493, NA, NA, NA, NA, NA, 103, NA, 202, NA, 10~
## $ Grade10 <dbl> NA, 442, NA, NA, NA, NA, NA, 86, NA, 177, NA, 101~
## $ Grade11 <dbl> NA, 390, NA, NA, NA, NA, NA, 87, NA, 139, NA, 96,~
## $ Grade12 <dbl> NA, 387, NA, NA, NA, NA, NA, 85, NA, 156, NA, 92,~
## $ Grade13 <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ Ungrade <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ MaleEnrollment <dbl> 459, 868, 451, 463, 304, 498, 192, 350, 222, 351,~
## $ FemEnrollment <dbl> 431, 844, 423, 411, 302, 446, 172, 314, 239, 323,~
## $ TotalEnrollment <dbl> 890, 1712, 874, 874, 606, 944, 364, 664, 461, 674~
## $ StudentTeacherRatio <dbl> 19.78, 20.09, 20.33, 20.33, 23.31, 15.48, 16.52, ~
## $ AIANMale <dbl> 4, 0, 1, 0, 1, 9, 1, 1, 3, 1, 0, 1, 0, 1, 3, 4, 0~
## $ AIANFemale <dbl> 1, 2, 4, 4, 3, 1, 3, 1, 3, 3, 0, 1, 0, 0, 1, 2, 2~
## $ AIANTotal <dbl> 5, 2, 5, 4, 4, 10, 4, 2, 6, 4, 0, 2, 0, 1, 4, 6, ~
## $ AsianMale <dbl> 4, 4, 4, 1, 2, 3, 0, 1, 0, 0, 4, 1, 0, 0, 2, 0, 1~
## $ AsianFemale <dbl> 2, 5, 0, 3, 0, 0, 0, 1, 0, 0, 0, 4, 0, 0, 3, 3, 1~
## $ AsianTotal <dbl> 6, 9, 4, 4, 2, 3, 0, 2, 0, 0, 4, 5, 0, 0, 5, 3, 2~
## $ BlackMale <dbl> 15, 23, 22, 22, 26, 24, 2, 8, 4, 6, 2, 3, 0, 7, 7~
## $ BlackFemale <dbl> 14, 34, 28, 16, 23, 21, 0, 3, 5, 4, 1, 2, 0, 5, 3~
```

```
## $ BlackTotal      <dbl> 29, 57, 50, 38, 49, 45, 2, 11, 9, 10, 3, 5, 0, 12~
## $ HPIMale         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0~
## $ HPIFemale       <dbl> 1, 0, 0, 0, 0, 1, 0, 2, 1, 0, 0, 0, 0, 0, 0, 0~
## $ HPITotal        <dbl> 1, 0, 0, 0, 0, 1, 0, 2, 1, 2, 0, 0, 0, 0, 0, 0~
## $ HispanicMale    <dbl> 251, 490, 263, 261, 167, 290, 6, 162, 92, 153, 2, ~
## $ HispanicFemale  <dbl> 251, 468, 241, 236, 152, 256, 8, 156, 102, 140, 5~
## $ HispanicTotal   <dbl> 502, 958, 504, 497, 319, 546, 14, 318, 194, 293, ~
## $ TRMale          <dbl> 17, 26, 7, 11, 4, 9, 5, 2, 2, 1, 0, 4, 4, 1, 4, 3~
## $ TRFemale        <dbl> 15, 19, 6, 16, 4, 10, 9, 3, 3, 2, 3, 4, 3, 3, 4, ~
## $ TRTotal         <dbl> 32, 45, 13, 27, 8, 19, 14, 5, 5, 3, 3, 8, 7, 4, 8~
## $ WhiteMale       <dbl> 168, 325, 154, 168, 104, 163, 178, 176, 121, 188, ~
## $ WhiteFemale     <dbl> 147, 316, 144, 136, 120, 157, 152, 148, 125, 174, ~
## $ WhiteTotal      <dbl> 315, 641, 298, 304, 224, 320, 330, 324, 246, 362, ~
## $ Latitude        <dbl> 34.26020, 34.26220, 34.27330, 34.25270, 34.28980, ~
## $ Longitude       <dbl> -86.20620, -86.20490, -86.22010, -86.22181, -86.1~
```

## Finding How Many Students Eligible for FRPL

```
sum(df0$LunchRate > 75)
```

```
## [1] 9942
```

```
sum(df0$LunchRate < 25)
```

```
## [1] 3804
```

```
tapply(df0$LunchRate > 75, df0$Locale, sum, na.rm = TRUE)
```

|    |                     |                   |                   |                  |
|----|---------------------|-------------------|-------------------|------------------|
| ## | 11-City: Large      | 12-City: Mid-size | 13-City: Small    | 21-Suburb: Large |
| ## | 2242                | 801               | 617               | 2362             |
| ## | 22-Suburb: Mid-size | 23-Suburb: Small  | 31-Town: Fringe   | 32-Town: Distant |
| ## | 212                 | 189               | 157               | 557              |
| ## | 33-Town: Remote     | 41-Rural: Fringe  | 42-Rural: Distant | 43-Rural: Remote |
| ## | 567                 | 798               | 702               | 738              |

```
tapply(df0$LunchRate < 25, df0$Locale, sum, na.rm = TRUE)
```

|    |                     |                   |                   |                  |
|----|---------------------|-------------------|-------------------|------------------|
| ## | 11-City: Large      | 12-City: Mid-size | 13-City: Small    | 21-Suburb: Large |
| ## | 476                 | 226               | 335               | 1490             |
| ## | 22-Suburb: Mid-size | 23-Suburb: Small  | 31-Town: Fringe   | 32-Town: Distant |
| ## | 133                 | 70                | 104               | 51               |
| ## | 33-Town: Remote     | 41-Rural: Fringe  | 42-Rural: Distant | 43-Rural: Remote |
| ## | 47                  | 530               | 186               | 156              |

```
sum(df0$BlackTotal[df0$LunchRate > 75], na.rm = TRUE)
```

```
## [1] 1225186
```

```
sum(df0$HispanicTotal[df0$LunchRate > 75], na.rm = TRUE)
```

```
## [1] 2184057
```

```
sum(df0$WhiteTotal[df0$LunchRate > 75], na.rm = TRUE)
```

```
## [1] 864627
```

```
sum(df0$BlackTotal[df0$LunchRate < 25], na.rm = TRUE)
```

```
## [1] 119086
```

```
sum(df0$HispanicTotal[df0$LunchRate < 25], na.rm = TRUE)
```

```
## [1] 375153
```

```
sum(df0$WhiteTotal[df0$LunchRate < 25], na.rm = TRUE)
```

```
## [1] 1496745
```

## Graphs

Faceted by Race (df0)

```
df0$predominantRace <- sapply(1:nrow(df0), function(row){
  race <- c("Black", "White", "Hispanic")
  totals <- c(df0$BlackTotal[row], df0$WhiteTotal[row], df0$HispanicTotal[row])
  if (all(is.na(totals)) || all(totals == 0)){
    return(NA)
  }else{
    return(race[which.max(totals)])
  }
})
df0 <- df0 %>%
  filter(!is.na(predominantRace) & predominantRace %in% c("Black", "White", "Hispanic"))
```

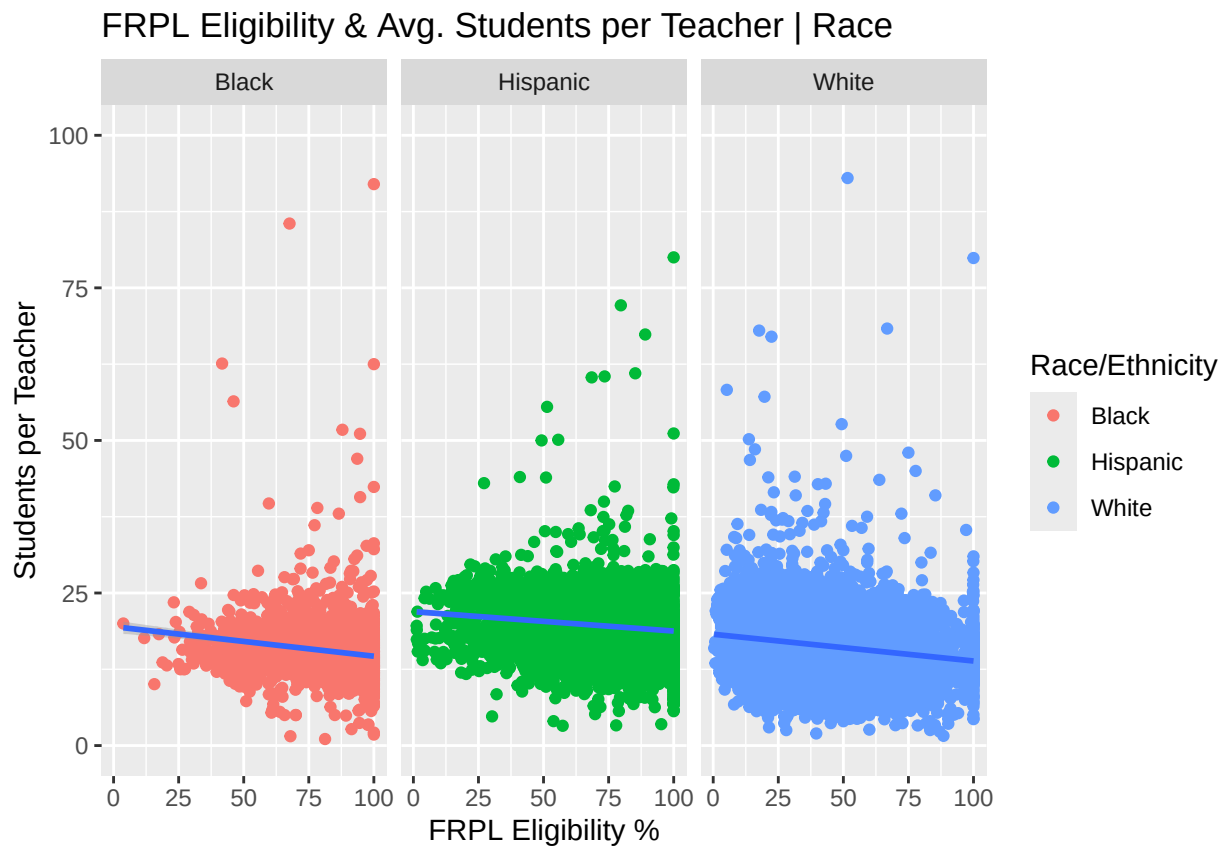
```
df0_sample <- df0 %>% sample_n(15000)
```

```
ggplot(df0_sample, aes(x = LunchRate, y = StudentTeacherRatio)) +
  geom_point(aes(color = predominantRace)) +
  geom_smooth(method = "lm") +
  facet_wrap(~predominantRace) +
  labs(title = "FRPL Eligibility & Avg. Students per Teacher | Race",
       x = "FRPL Eligibility %",
       y = "Students per Teacher",
       color = "Race/Ethnicity") +
  scale_y_continuous(limits = c(0, 100))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 12 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 12 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
### Faceted by Race (df_all)
```

```
df_all$predominantRace <- sapply(1:nrow(df_all), function(row){
  race <- c("Black", "White", "Hispanic")
  totals <- c(df_all$BlackTotal[row], df_all$WhiteTotal[row], df_all$HispanicTotal[row])
  if (all(is.na(totals)) || all(totals == 0)){
    return(NA)
  }else{
    return(race[which.max(totals)])
  }
})
df_all <- df_all %>%
  filter(!is.na(predominantRace) & predominantRace %in% c("Black", "White", "Hispanic"))
```

```
df_all_sample <- df_all %>% sample_n(15000)
```

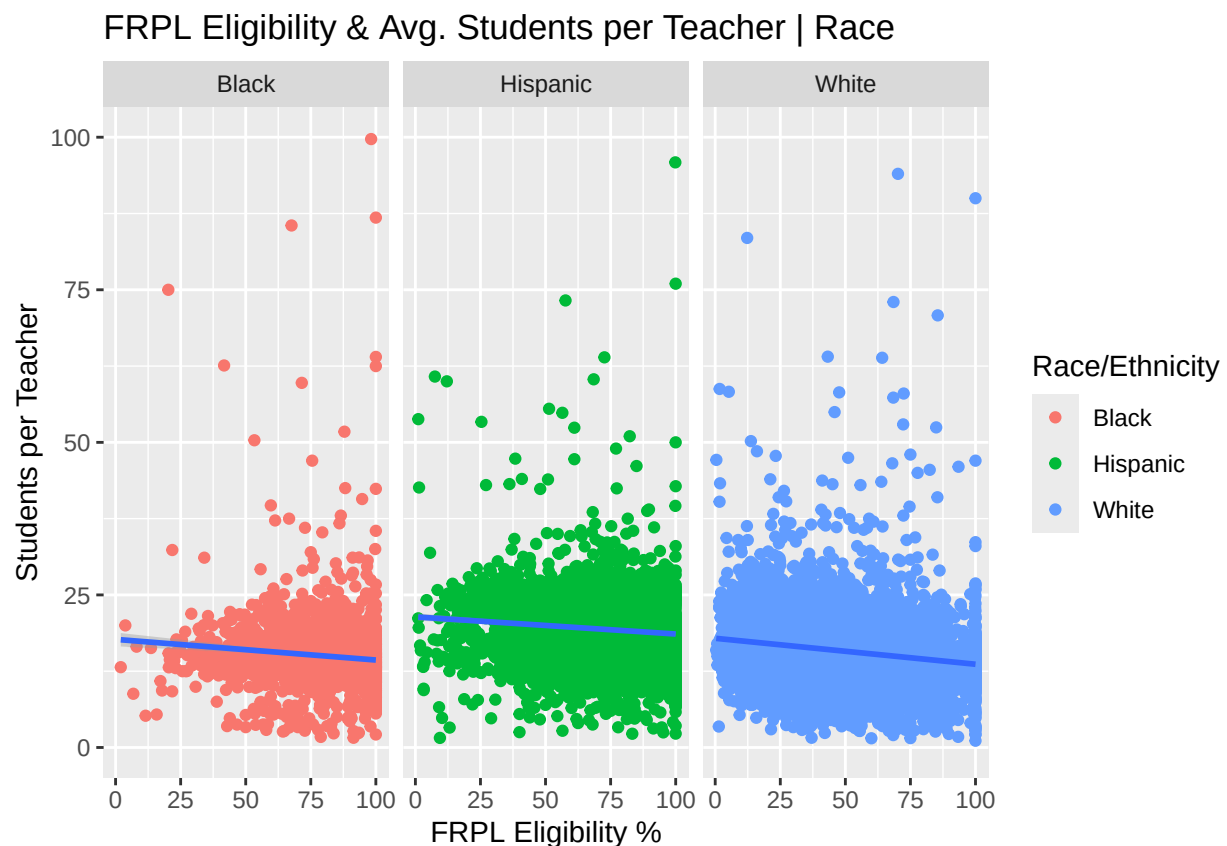
```
ggplot(df_all_sample, aes(x = LunchRate, y = StudentTeacherRatio)) +
```

```
geom_point(aes(color = predominantRace)) +
geom_smooth(method = "lm") +
facet_wrap(~predominantRace) +
labs(title = "FRPL Eligibility & Avg. Students per Teacher | Race",
      x = "FRPL Eligibility %",
      y = "Students per Teacher",
      color = "Race/Ethnicity") +
scale_y_continuous(limits = c(0, 100))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 14 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_point()').
```



## Locale Graph

```
df0$locale <- factor(df0$Locale, levels = c(
  "11-City: Large", "12-City: Mid-size", "13-City: Small",
  "21-Suburb: Large", "22-Suburb: Mid-size", "23-Suburb: Small",
```

```

"31-Town: Fringe", "32-Town: Distant", "33-Town: Remote",
"41-Rural: Fringe", "42-Rural: Distant", "43-Rural: Remote"
))

locale_sample <- df0 %>% sample_n(3000)

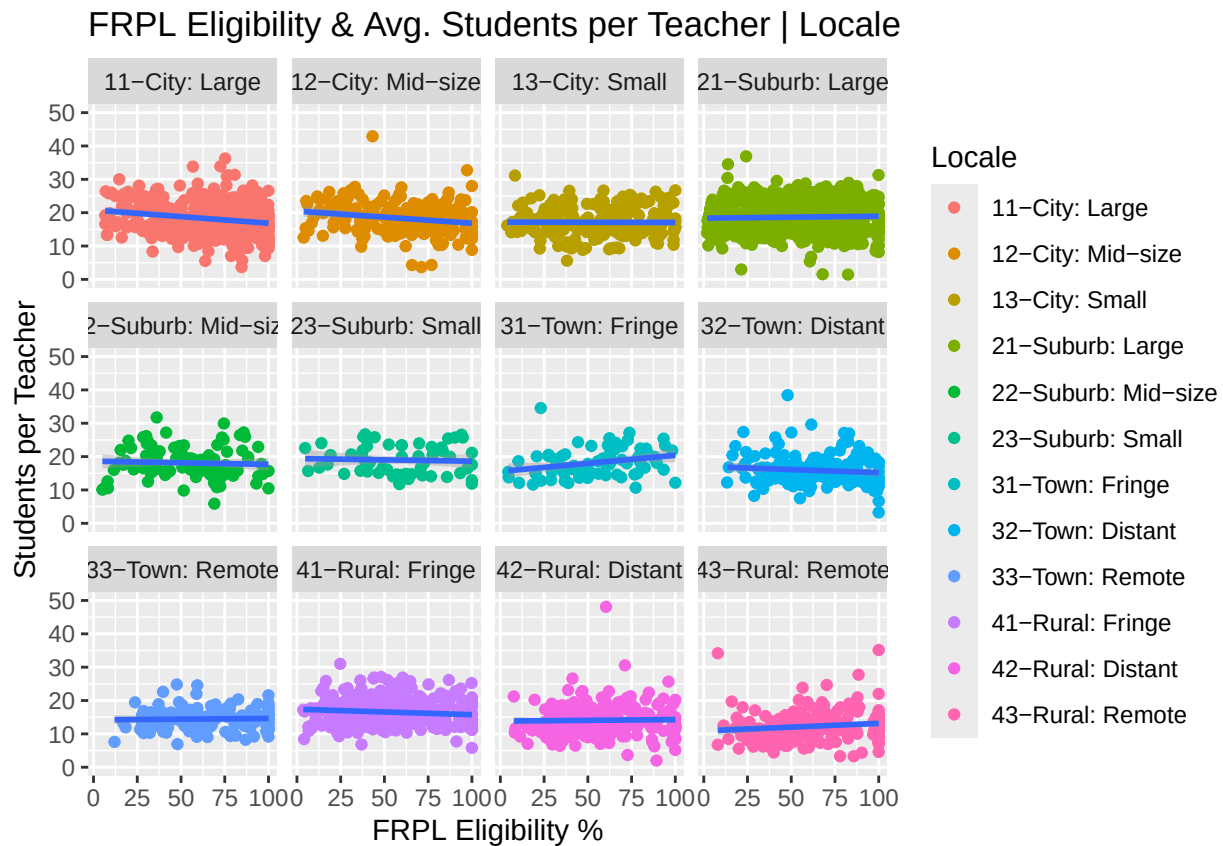
ggplot(locale_sample, aes(x = LunchRate, y = StudentTeacherRatio)) +
  geom_point(aes(color = Locale)) +
  geom_smooth(method = "lm") +
  facet_wrap(~locale) +
  labs(title = "FRPL Eligibility & Avg. Students per Teacher | Locale",
       x = "FRPL Eligibility %",
       y = "Students per Teacher") +
  scale_y_continuous(limits = c(0, 50))

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 7 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 7 rows containing missing values or values outside the scale range
## ('geom_point()').

```



## Tests

### Correlation

```
cor.test(df0$LunchRate, df0$StudentTeacherRatio)

##
## Pearson's product-moment correlation
##
## data: df0$LunchRate and df0$StudentTeacherRatio
## t = -2.8527, df = 30035, p-value = 0.004338
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.027761856 -0.005150045
## sample estimates:
## cor
## -0.01645805
```

### MANOVA

```
manova_result <- manova(cbind(LunchRate, StudentTeacherRatio) ~ predominantRace + Locale, data = df0)
summary(manova_result)
```

```
##              Df  Pillai approx F num Df den Df    Pr(>F)
## predominantRace    2 0.32807   2945.55     4 60046 < 2.2e-16 ***
## Locale             11 0.09197    131.57    22 60046 < 2.2e-16 ***
## Residuals          30023
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary.aov(manova_result)
```

```
## Response LunchRate :
##              Df   Sum Sq Mean Sq F value    Pr(>F)
## predominantRace    2  6223416 3111708 6833.29 < 2.2e-16 ***
## Locale             11  1238169  112561  247.18 < 2.2e-16 ***
## Residuals          30023 13671715     455
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response StudentTeacherRatio :
##              Df   Sum Sq Mean Sq F value    Pr(>F)
## predominantRace    2   95328   47664 238.031 < 2.2e-16 ***
## Locale             11   71101    6464  32.279 < 2.2e-16 ***
## Residuals          30023 6011907     200
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



## 1.8

The correlation test suggests a very small negative relationship between the percentage of FRPL-eligible students and class size, meaning that as the FRPL-eligible percentage increases, the number of students per teacher declines very slightly. For the MANOVA test using LunchRate and StudentTeacherRatio as dependent variables, the predominantRace variable has a more significant effect on the dependent variables than locale, based on the Pillai values.

## 1.9

The research question explores whether there is a correlation between free/reduced lunch rates and student-to-teacher ratios in traditional U.S public schools, and if it reveals resource disparities based on students' racial or ethnic backgrounds and school geography? Starting with the plots faceted by race, there is a decline in students per class as the FRPL eligibility increases in both dataframes. This is also confirmed by the correlation test. For the plot faceted by locale, there is a bit more variation; most locales seem to also have a decline in students per class as FRPL eligibility increases, 31-Town and 32-Town show a slight incline however. Suburban schools seem to have virtually no incline or decline. These trends are confirmed by the MANOVA test. Though these results contradict with my theory that class sizes would actually increase as FRPL eligibility increases, it does reveal interesting notes about which schools are more likely to have FRPL eligible students and might experience outliers in class sizes.

## 1.10

A significant limitation that exists as a result of this analysis is that resource allocation is not solely represented by class sizes, and that there are other indicators that are not included in this study.

## Sources

```
#https://r-coder.com/tapply-r/  
#https://www.spsanderson.com/steveondata/posts/2024-08-26/#introduction  
#https://r-graph-gallery.com/223-faceting-with-ggplot2.html  
#https://www.youtube.com/watch?v=CshEAjYQN8U
```